Lorenzo Di Napoli

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Personal Introduction

Student of Engineering of Computing Systems at Politecnico di Milano (Grade Average: 26.11/30), with skills in low-level programming (*VHDL · Assembly Risc-V · C · C++ · CUDA*) and interest in embedded systems and robotics. Focused on understanding systems deeply and the influence of hardware architecture on software applications.

Educational Path

2023 - expected 2026: Bachelor's Degree in Engineering of Computing Systems, Politecnico di Milano (MI) – Grade Average: 26.11/30.

Extracurricular Courses: GPU Architecture (CUDA) · FPGA Architecture (VHDL & HLS)

2018 - 2023: High School Diploma, in Liceo Scientifico opzione Scienze Applicate, Istituto Salesiano Sant'Ambrogio, Milano (MI).

Technical Skills & Soft Skills

Programming Languages

C/C++	● ● ● ● ●	**Cuda**	● ● ○ ○ ○	**Assembly**	● ● ● ○ ○
Java	● ● ● ● ○	**Python**	● ● ○ ○ ○	**VHDL**	● ● ● ○ ○

Tools and Support Technologies

- Linux (Ubuntu, Fedora)
- Makefile · CMake
- Docker/Podman
- Git · GitHub
- Valgrind · Cachegrind · Massif · GDB
- Vivado · Vitis

Soft Skills from Work Experiences

- Effective Communication and Teamwork
- Management of Emergencies and Decision-Making under Pressure
- Accountability for own and others' safety and legal responsibilities
- Supervision and Management of Customer Safety
- First Aid and Defibrillator (BLSD) Certification

Technical Knowledge from University Courses

Computer Architectures (RISC-V) · Digital Logic Design · Operating Systems (Linux)
Algorithms and Principles of Computer Science · Software Engineering
Fundamentals of Internet and Communication Networks · Database and Information Systems
Basic Circuit Theory · Introduction to Electronics · Fundamentals of Automatic Control

GPU Architecture (CUDA) · FPGA Architecture (VHDL & HLS)

Work Experiences and Projects

University Projects

Movhex: development in **C** of a program that allows creating and editing a 2D hexagonal map, with the objective of finding the lowest-cost path between two points.
The final result reached a temporal complexity of $O(n\log(n))$ and spatial complexity of $O(n)$; where n is the number of hexagons present in the map.

Priority-List: design and implementation in **VHDL** of a HW module, which performs the function of a controller for a Priority-List of Tasks.
The final result was implemented using a fully synthesizable FSM on an FPGA without Latches.

Personal Projects

LDN Library: personal general-purpose library implemented in **C++20** using the advanced features of the language.

Neural Networks: from-scratch implementation of different neural networks in Python, focused on clustering various geometric shapes (circles, spirals, etc.).

Work Experiences

Lifeguard: First aid and defibrillator (BLSD) certification, civil and criminal liability for one's actions and those of swimmers.

- **June 2025 - December 2025 [116 hours]:** Summer and winter work experience as Lifeguard Assistant P at the sports center *"M.G.M.Sport Srl"* in Paderno Dugnano (MI).
- **June 2024 - July 2024 [171 hours]:** Summer work experience as Lifeguard Assistant P at the sports center *"M.G.M.Sport Srl"* in Paderno Dugnano (MI).
- **July 2022 [36.5 hours]:** Summer work experience as Lifeguard Assistant P at the sports center *"M.G.M. Sport Srl"* in Desio (MB).

Language Skills

Italian: Native speaker.

English: Level C1, Grade C (Cambridge CAE Certificate).

Certifications and Documents —> DOWNLOAD

Driver's License: Category B driver's license — [18/11/2023]

Cambridge CAE Certificate: English language certification, level C1 Grade C — [09/01/2026]

Lifeguard Certification: award of the Lifeguard Assistant certificate (levels P, IP, MIP) at the *"Federazione Italiana Nuoto" (FIN)* — [09/11/2022]

SFA Certificate: certificate of participation and completion of the *"Scuola Formazione Animatori MGS" (SFA)* at the association *"Movimento Giovanile Salesiano" (MGS)* — [17/04/2023]